Acute Heat Stress and Cranberry Yield

Executive Summary

Introduction

Cranberry production holds a major economic and cultural significance in Wisconsin and the United States. Cranberries require particular conditions to grow and thrive. Wisconsin happens to provide suitable temperature and conditions for production of fruit. There is a growing necessity to investigate the effect of raising global temperature on the production of cranberries. This study presents the results of acute heat stress experiments across 2024 and 2025 which found that heat treatment reduced both total healthy fruit weight.

Findings

The study was conducted on Stevens(ST) and Mullica Queen(MQ) cultivar plots with each three instances of an untreated plot paired with a heat treated plot (4-1 pulse(s)). Each heat shock lasted 2 hours. Therefore the study consisted of 48 paired plots for each cultivar (96 total).

Across all intensities, years and cultivar, heat treatment reduced total fruit weight by ~12% (95% Confidence Interval(CI):[7.4%-16.5%]) and reduced healthy weight by ~20% (95% CI: [14.2%, 25.0%]) roughly between 14-25%). The largest estimated losses were in 2025 where the four-pulse treatment was about 37% (95% CI: [28.1%, 45.5%]) below control for total weight and about 40% below for healthy weight. The average loss differed by year with 15% (95% CI:[9.8%, 20.7%]) in 2025 as opposed to 8%(95% CI: [3.2%, 12.8%]) in 2024 for total weight. Healthy fruit count and weight proportions dropped by 11%(95% CI:[4.1%,10.3%]) and 7%(95% CI: [4.1%, 10.3%]) respectively, pointing to a higher incidence of fruit rot. Across the four intensity levels, total weight fell by about 9%(95% CI:[5.1%,12.3%]) per heat pulse overall. Neither the ST nor the MQ showed sufficient statistical evidence in response to heating, although ST was seen with lower levels of cranberry production.

Heat treatment reduced healthy fruit weight more than total fruit weight, meaning that total harvest weight alone understated the loss of healthy fruit. For involved stakeholders, these findings support tracking healthy fruit weight and rot fruit share alongside total yield when assessing heat damage.

Final Reflections and Suggestions

As heat pulses increased, it had a diminishing effect on the production of cranberries regardless of year or type of cultivar. We can therefore conclude that acute heat stress has a negative impact on total weight, healthy weight and healthy fruit count on cranberry production. There is a clear correlation between temperature and production capability/capacity of cranberries. Stakeholders are advised to be diligent with the concerns of raising temperatures in the midwest and adapt in whatever aspects of the production that addresses the production of cranberries.

Contributions

Author contributions are listed in the table below.

Author	**Contribution**	**Share of the work**
Soren Basnet	Technical Report, Presentation, Revision of Executive Summary	Section 5,6 from the technical report. Worked on the structure and portions of the presentation. Made corrections and contributions to executive summary.
Qingquan Liu	Technical Report, Presentation, Executive Summary	Section 1, 2 ,4 Worked on visualization and portions of the presentation.
Wuyang Wang	Technical Report, Executive Summary, Presentation	Report section 3 & appendix A. Worked on presentation. Fix bugs in Report. Draft the executive summary.

References

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AI Usage Statement

1) Tools : Chat GPT-6

Usage : Code revision, syntactical error correction and modeling

Initial prompt : " I want to perform beta-binomial regression, what is R code syntax for that ", "I want to diagnose logistic regression model, how do I diagnose it ? ", "How to display the output of said model in a manner that is presentable ? "

Responsibility : Despite using the aforementioned chatbot to write, improve and troubleshoot syntactical error involving R programming language, all the decision, initial draft code and validation for the script generated was performed by the team.

2) Tools : Gemini 3.1 Pro

Usage : Code revisions, graphs and visualization

Initial prompt : "I have this base R code to plot data distribution for variable X, how do I plot it in a visually appealing manner in R ? ", " How do I add my custom color to the jitter points in this R code ? "

Responsibility : No output of the said visualization was input into the chatbot. Only the code chucks were used as inputs for making improvements. Interpretation of the graphs and visualizations were made by the team.

3) Tools : Gemini 3.1 Pro

Usage : Theoretical and statistical interpretation

Initial prompt : " How do I understand the output of the beta-binomial model in R ? ", " What does a residual plot that is of this nature signify ? "

Responsibility : This task included understanding of models and the type of models suited for variable response variables being predicted. It was also used to learn about statistical models and theory for the purpose of the project.

4) Tools : Chat GPT-6

Usage : Grammatical error correction

Initial prompt : " I believe I am using the work "correlated" way too much in this paragraph, how to modify the sentence without losing meaning ? ", "Can you check for any grammatical errors ? "

Responsibility : All description and paragraphs were written by the team before making each paragraphs succinct and readable.

5) Tools : Zero-GPT (non-AI)

Usage : Check for AI usage percentage for executive summary

Initial prompt : "Check for AI written texts in this report "

Responsibility : Only about 21% of the texts in this document were highlighted as written by AI. We attribute this to the grammatical corrections and awkward sentence

corrections we made using AI. Not clear numerical output was mentioned in the version of the document used for checking for AI usage.

6)Data cleaning and clustering

Tool & version: Deepseek-V4.1-Flash_1017

Description: check our code

Initial prompt: Here is the initial code. We tried to group the data in following steps: 1.separate the data with year 2024 & 2025; 2.separate the data with ST and MQ in 2024 & 2025. 3. separate the pulse frequency. The control group's pulse is 0, It only depend on cultivars and year; 4.calculate the total weight part of each pulse an percentage of healthy fruit and count 5.set the variables. We now have trouble in set them as variables, please debug.

Responsibility: we overview the code and learn the meaning of thee code. We first viewed the tables and checked the structure. We also calculate some part by ourselves.

7) Data exploration

Tool & version: Deepseek-v4.1-Flash_1017 & GPT-6.1 sol

Description: We need to draw pictures to visualize the real data and use residual plots and correlation matrix to decide what kind of model we are ready to use.

Initial prompt: Check the code and our model selection below. We create a 1*2 preview plot of the real data. Each group's three data use the mean value. The plot is distinguished by year and cultivar. We draw the scatter plots of total weight, healthy weight, percentage of healthy weight and count. We draw a correlation matrix of each variables. If our graphs can interpret our model we decide tho use?

Responsibility: We confirmed the answers from AI and tried to improve our models

8) Figure styling

Tool & version: Claude Opus 5.5

Description: Used to improve the readability of the figures: axis and legend labels, colour, point size and transparency. We decided which messages the figure should convey and checked the output.

Initial prompt: "This is my code. How can I improve the output?"

Responsibility: We ran all plotting code on our data, checked that each figure matches the underlying numbers, and checked every caption against the figure.

9) Model choice

Tool & version: Claude Sonnet 5.5

Description: Used to discuss which statistical model suits our data. It suggested we can try a linear mixed mode。We chose the final model structure ourselves based on our exploratory plots, the instructor's guidance and model comparison.

Initial prompt: "Which statistical model is suitable for this data?"

Responsibility: We fitted all models in R ourselves, compared candidate models, checked the residual diagnostics, and checked that our conclusions follow from our own output.

10) Debugging PDF compilation

Tool & version: Claude Opus 5.5

Description: Used to identify and fix a PDF compilation error caused by stray \n characters in the merged R Markdown file.

Initial prompt: "Why can't the PDF be generated?" (followed by the LaTeX error message)

Responsibility: We checked the suggested corrections and confirmed that the final document could be successfully knitted to PDF without errors.